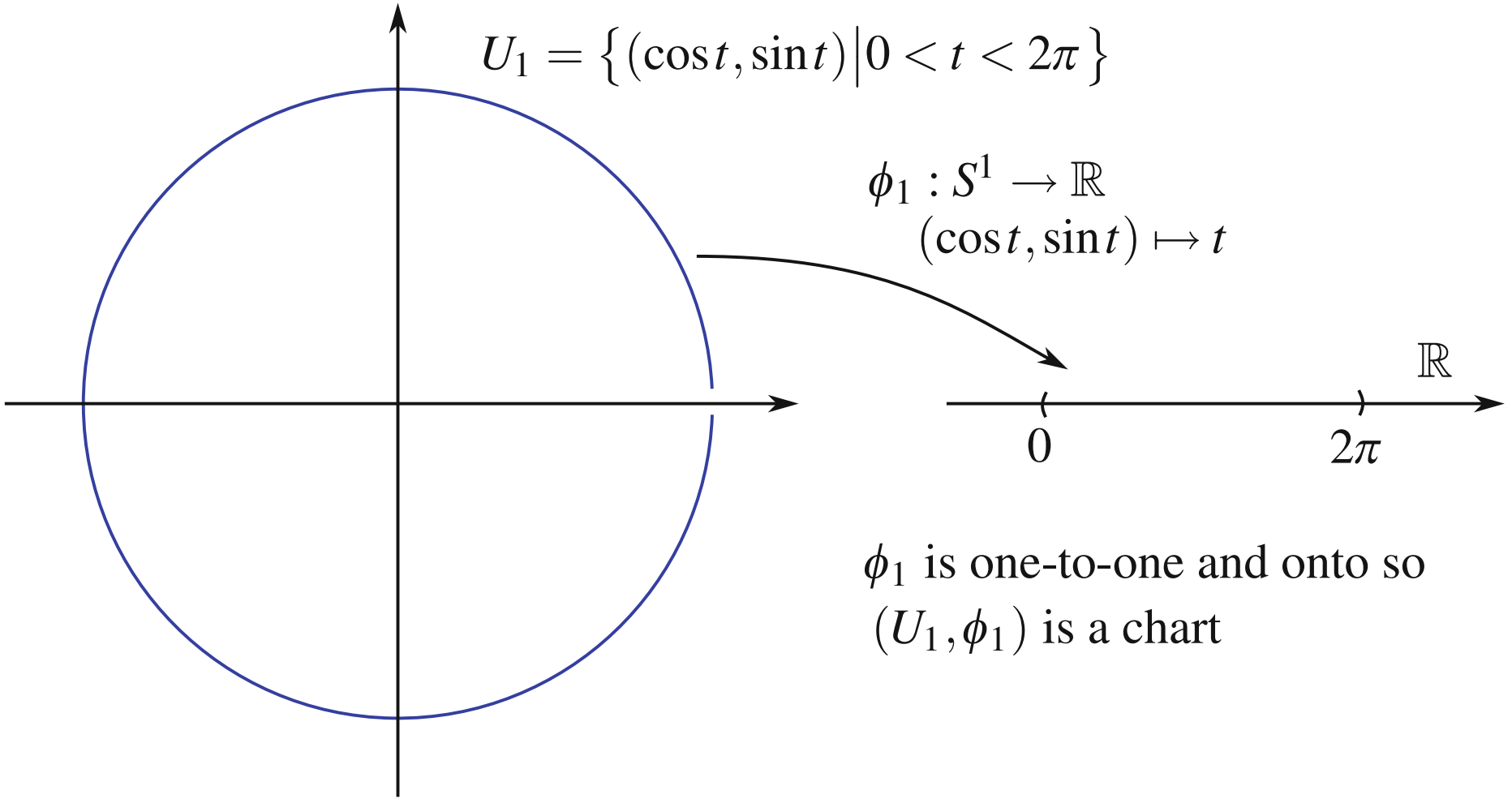

$U_1 = \{(\cos t, \sin t) \mid 0 < t < 2\pi\}$
$\phi_1 : S^1 \to \mathbb{R}$
$(\cos t, \sin t) \mapsto t$
$\mathbb{R}$
0
2π
ϕ_1 is one-to-one and onto so
(U_1, ϕ_1) is a chart